

Degree Project in the Field of Technology Media Technology and the Main Field of Study Computer Science and Engineering

Second cycle, 30 credits

Design Guidelines for Simulated Air Combat Planning Tools

Supporting high learnability and low error rates

VIKTORIA BJÖRKLUND

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Master's Programme, Interactive Media Technology, 120 credits
Date: June 22, 2026

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Swedish title: Designriktlinjer för planeringsverktyg för simulerade luftstrider
Swedish subtitle: Designval för hög inlärningsförmåga och låga felfrekvenser

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Military air combat training relies on complex scenario planning, where usability of planning tools is critical for efficiency and accuracy. This study explores which interface design choices are associated with high learnability and low error rates in a map-based military planning tool. A web-based prototype was developed using React and Leaflet, and evaluated through user testing with six participants, including two from the intended user group. Learnability was assessed by comparing task completion times across repeated tasks, while error rate was measured through incorrect interaction attempts. The results indicate that consistent interaction patterns and direct manipulation of map elements support both rapid learning and low error rates. In contrast, more complex and less visible interactions, such as creating geometric shapes and formations, led to increased errors and user uncertainty. The findings also suggest that prior experience influences both performance and perceived usability, highlighting limitations in using error rate as a standalone metric. Overall, the study provides design guidelines for improving usability in map-based planning tools within similar domains. These insights contribute to the design of more accessible and efficient planning tools for complex operational environments.

SAMMANFATTNING

Militär luftstridsträning bygger på komplex scenarioplanering, där användbarheten av planeringsverktyg är avgörande för effektivitet och tillförlitlighet. Denna studie undersöker vilka gränssnittsdesignval som är förknippade med hög inlärningsbarhet och låga felfrekvenser i ett kartbaserat militärt planeringsverktyg. En webbaserad prototyp utvecklades med hjälp av React och Leaflet och utvärderades genom användartester med sex deltagare, varav två från den avsedda användargruppen. Inlärningsbarheten bedömdes genom att jämföra slutförandetider för upprepade uppgifter, medan felfrekvensen mättes genom felaktiga interaktionsförsök. Resultaten indikerar att konsekventa interaktionsmönster och direkt manipulation av kartelement stöder både snabb inlärning och låga felfrekvenser. Däremot ledde mer komplexa och mindre synliga interaktioner, såsom att skapa geometriska former och formationer, till ökade fel och användarosäkerhet. Resultaten tyder också på att tidigare erfarenhet påverkar både prestanda och upplevd användbarheten, vilket belyser begränsningar i att använda felfrekvens som ett fristående mått. Sammantaget ger studien designriktlinjer för att förbättra användbarheten i kartbaserade planeringsverktyg inom liknande områden. Dessa insikter bidrar till utformningen av mer tillgängliga och effektiva planeringsverktyg för komplexa operativa miljöer.

CCS Concepts: • **Human-centered computing** → **Interactive user interfaces**.

Keywords: human-computer interaction, usability, learnability, military planning tools, simulated air combat, map-based interfaces, direct manipulation

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Document date: June 22, 2026

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Nyckelord: människa-datorinteraktion, användbarhet, inlärningsbarhet, militära planeringsverktyg, simulerad luftstrid, kartbaserade gränssnitt, direkt manipulation

1 INTRODUCTION

At the Swedish Air Force's air combat simulation center (FLSC), Air Force pilots and air combat controllers practice combat in large, complex scenarios that include both friendly and opposing units. The scenarios are carefully planned before the exercise begins and include, among other things, the geographical placement and configuration of units. The scenarios are both time-consuming and resource-intensive, making the quality of the planning tool critical. Poorly designed or inefficient planning processes risk reducing the overall training value of the exercises.

The current planning tool used at FLSC is characterized by a complex interface and is restricted to on-site use only. As a result, only a limited number of individuals have developed sufficient proficiency to operate it flawlessly. Furthermore, since the system is only accessible at a specific physical location, other members of the intended user group are unable to plan upcoming scenarios with the tool beforehand. The new planning tool aimed to address these limitations by redesigning the interface to improve usability and enable access independently of geographical location, thereby supporting a larger portion of the intended user group.

Similar to many other military systems, the existing planning tool features a complex interface with numerous functions and interaction elements. To evaluate whether the redesigned planning tool improves ease of use and therefore allows more intended users to operate it efficiently, learnability and error rate were selected as primary evaluation metrics. Learnability was measured by completion time between tasks of the same character, and error rate was captured by the number of failed interface navigation actions.

In order to investigate this, a prototype was implemented in JavaScript using both the Leaflet and React libraries. The prototype was evaluated with six participants through user tests with follow-up interviews afterwards. The study aims to contribute guidance on which design choices can be made within similar systems to support ease of use. The research question guiding the study is:

"Which design choices are most critical for supporting low error rate and high learnability in a digital map-based planning tool for air combat scenarios?"

2 BACKGROUND

This section explains the operational context, the nature of the planning tool and the meaning of learnability and error rate.

2.1 Features of the planning tool

The main features of the current planning tool include the placement of units, start and landing bases, waypoints, reference points, polygons, circles, and aircraft (see Figure 1). Start bases represent where aircraft begin their mission, while landing bases indicate where they return after completed operations. Waypoints are geographical points that define the intended flight route and contain associated information such as altitude and speed. Reference points primarily serve as navigational guidance and support orientation in unfamiliar terrain.

Polygons and circles are used to represent areas or conditions relevant to the scenario rather than route planning itself, providing pilots with additional contextual information about the operational environment. All entities contain associated geographical coordinates, including latitude and longitude.

The planning tool is used prior to the execution of air combat scenarios and is therefore not operated under immediate time pressure. Its primary users are military staff members who, based on requests and input from visiting fighter pilots, configure and position scenario elements within the system. The map constitutes the central interaction component of the interface and supports accurate spatial placement of units and other scenario elements.

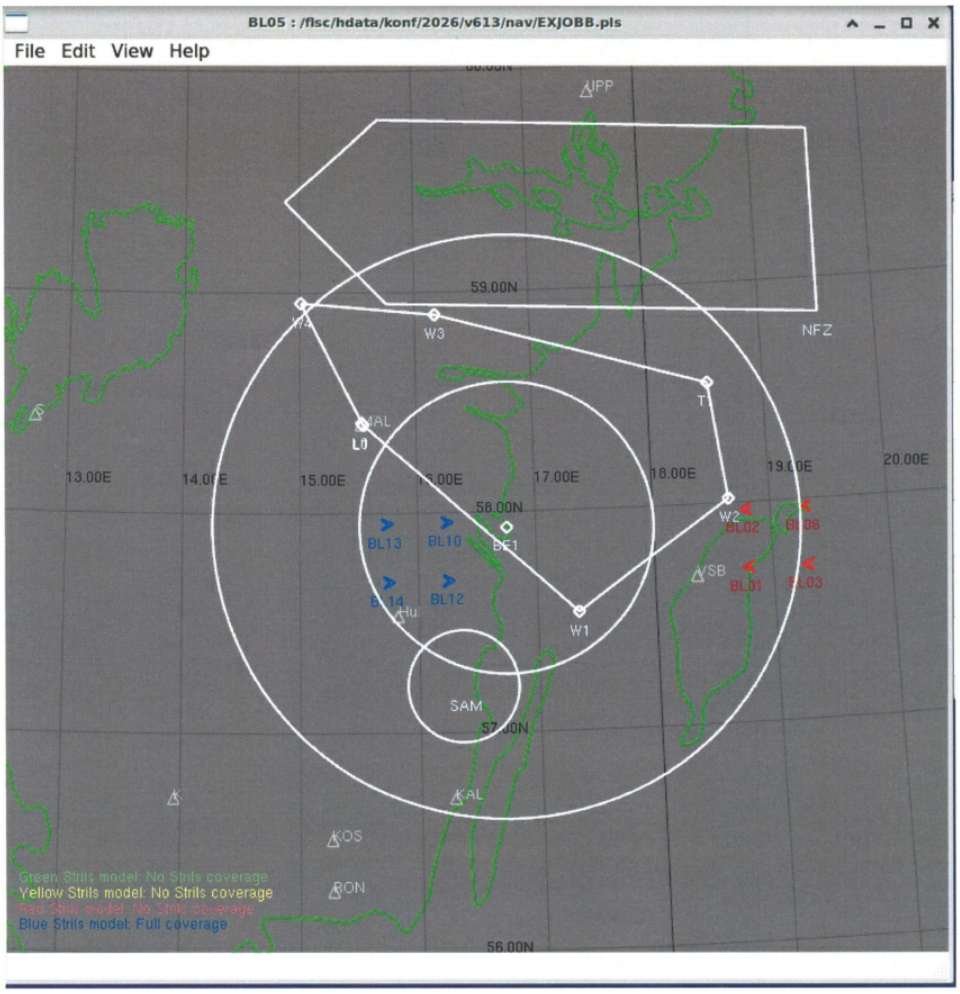


Figure 1: The planning tool that is currently used.

2.2 Learnability and Error rate

2.2.1 Definition. The definition of usability is often associated with five attributes: Learnability, Efficiency, Memorability, Errors and Satisfaction [17]. Learnability, the author argues, refer to that the system should be easy to learn in order to have the user rapidly starting to know how to navigate in the

system. Furthermore, the system should have a low error rate in order to have the users making few errors. Learnability can be evaluated by measuring the time required to successfully complete tasks, where a predefined threshold determines whether the user is considered to have learned the system. An error is defined as any action that does not accomplish the desired goal, and the error rate of a system is measured by counting the numbers of such actions made by the user when performing a task.

2.2.2 Rationale for Using Learnability and Error Rate. The training required for users to achieve and maintain competence constitutes a significant cost of system operation [9]. The user interface can influence how much training is required by incorporating design properties that make the system easier or more difficult for users to learn. This speaks for the argument that learnability is a valid objective to measure, as it can save costs and time. The number of errors is highly related to changes in cognitive workload and has the potential to increase it [13]. Cognitive workload is described as the mental effort required to complete a task, and it has been shown that increased cognitive demand impaired task performance ([23],[3]). In the first study, it was revealed via tools such as IMPRINT, that cognitive workload can be quantitatively modeled and evaluated in military contexts. This supports the argument that cognitive demand is a relevant measurement to take into consideration, thus supporting the use of error rate as a behavioral indicator potentially associated with cognitive demand.

2.3 Maps for decision making

2.3.1 Effects on situational awareness. Situational awareness is described as *"the perception of one's environment and the ability to make informed predictions based on that perception"*, which is used in many different fields, including aviation, air traffic control and the military [15]. Lack of situational awareness increases the probability of human error, making it important for safety and performance [5]. Visualization of terrain and tactical data increased situational awareness in a case study covering a large geographic area containing many military units [7]. In another study, it is claimed that situational awareness is particular important for military and Air Force and regarded as the decisive factor in air combat engagements [16]. The capability to model and analyze a Geographic Information System (GIS) provide military commanders the opportunity to rapidly plan strategies and evaluate options [8]. Although the planning process does not involve real-time air combat execution, users still require a high level of understanding of the operational situation in order to configure meaningful training scenarios. Misunderstandings in the interface may therefore reduce the training value of the resulting scenarios, which supports the use of a map-based interface.

2.3.2 Interface Design and Cognitive Demand. In order to support and assist military commanders in their job of successfully carrying out and correctly perceiving numerous high-level tasks, proper visual user interfaces are needed [25]. Consistent interface structures and interaction mechanisms can reduce the mental effort required to operate a system, enabling users to devote more attention to operational decisions. Conversely, interfaces that conflict with user expectations may increase the likelihood of human error. [11]. This is supported by [14] who argue that consistency may be important for difficult tasks with a high cognitive load. In a study that compared a number of tools involving different degrees of detail and ways of spatial representation of the information, it was shown that the cognitive effort can change with different approaches for presenting output information [1]. In addition, it was found that the map that provided most information was not considered most useful by the participants and did not produce the best results.

2.3.3 Direct manipulation as a design. *"Direct manipulation is an interaction style in which UI elements are visible and can be acted upon via actions that receive immediate feedback"* [21]. Furthermore, direct manipulation covers a wide range of use cases, such as using your fingertips to pinch in and out in order to zoom on a picture. Allowing direct manipulation such as for accessing and moving data makes it possible to design systems that allow novice users to learn basic functionality rapidly and let them see if their actions are helping them reach their goals [22]. In map-based systems, direct manipulation may be particularly beneficial, since users can interact with spatial objects directly within their geographical context. This reduces the need to shift attention between separate interface components and may therefore lower the cognitive demand. Interfaces based on direct manipulation and consistent interaction metaphors can improve learnability by producing smoother and faster-growing learning curves compared to menu-based interfaces [20].

2.4 Related work

In a previous study, the authors explored system learnability through a layered interface design pattern deployed in a military mission planner interface that was evaluated with military personnel [9]. This pattern was used in order to achieve progressive disclosure, another interaction design pattern that limits the information and features presented at one time in order to make the interface less error-prone and facilitate for users. Data trends from the study indicated that multi-layered interfaces quickly can teach a user how to navigate through a complex user interface when first learning it, as well as preventing errors.

In another study, two alternative interfaces developed for military command and control were compared, where one (RAPTOR) followed ecological interface design (EID) and the other one (GGUI) was based on the army's Force XXI Battle Command Brigade and Below (FBCB2) interface [10]. Furthermore, the RAPTOR interface offered direct manipulation through a variety of control input mechanisms. Ecological interface design is a framework that focuses on the design of effective interfaces and displays, where two sets of mappings are critical. One relationship involves the mapping between the ecology and the informational content on the interface, assuring that critical information is present on the interface and that required control inputs are supported. The other one involves the mapping between the informational formats in the interface and the domain practitioner, assuring that information can be picked up easily and that the control surfaces allow inputs to be executed efficiently. The second interface (GGUI) produces multiple windows that contain relevant information (e.g, the combat resources of a specific unit) when the user points and clicks on button and tabs. Three tactical scenarios were simulated, and findings from the study suggests that the RAPTOR interface was easy to learn and use. Furthermore, subjective workload was significantly lower with the RAPTOR interface than with the GGUI, and reports were completed with the RAPTOR interface with approximately twice the accuracy (100% vs. 50%) relative to the GGUI.

Even highly detailed features have impacts on the learnability. When investigating icon type, icon size, and map background in a military GIS interface, it was found that an icon size of 80 min of the visual angle with an observation distance of 50 cm and a 21-inch screen produced optimum learning performance [24]. Walsh et.al suggest that using color according to relevant military standards could increase usability and decrease cognitive load [25]. Furthermore, they argue that compliance with military symbology standards would increase the usability of the products, as well as reduce cognitive load. In a study evaluating user experiences among military personnel in a multi-surface simulation training environment, the authors emphasized that visual feedback should be displayed at the user's point of

touch [4]. In a study where the author developed a simple geoportal, it was suggested that having polygons be overlaid other data without hiding any underlying information can be of advantage compared to symbolizing polygons with solid hatched fill [18]. Previous research indicates that users prefer direct map interaction for navigation, such as zooming and panning within the map interface, rather than using separate navigation controls. [19]. The authors also recommend minimizing the need for scrolling and keeping layer hierarchies shallow to improve overview and usability.

Although previous research has investigated layered interfaces, ecological interface design, and military GIS usability, limited research has explicitly connected specific map-based interface design choices to measurable learnability and error rate outcomes in air combat planning contexts.

3 METHOD

This section describes the methodology used to design, implement and evaluate a military planning tool used for air combat simulations. The development process followed an iterative design approach, involving exploration, prototyping, and evaluation [12]. Insights were gathered through demonstrated usage of the tools along with conducted semi-structured interviews with operators of both the current tool and similar ones, which eventually formed the user interface requirements. Evaluation was done through user tests with the product where task completion time and failed attempts were measured along with follow-up interviews.

3.1 Insights and desired features from operators

The initial phase of the study consisted of gathering data on three occasions in order to develop an understanding of the current problem. This was done by having the operators of the current tool demonstrate current usage on site, as well as highlighting desired features in a new planning tool. Written notes were taken as a foundation for later steps where design requirements would be defined in consultation with the operators who are part of the end users. An extensive literature review was conducted in order to form design guidelines based on what similar systems have implemented, and what design choices that seems to work well. A trip to one of the air flotillas in Sweden was conducted in order to gain understanding of the usage and logic of a similar tool used for flight route planning. An operator held a demonstration of the tool and explained the logic's of the available operations and of the interface logic.

3.2 Design process

The design process was characterized by iterative design cycles. During these cycles, the map prototype was presented to the operators, who provided feedback that was subsequently used to refine the design. The map was created in QGIS and later exported to Leaflet code for implementation in the web-based prototype. The menu components of the prototype were initially designed as a low-fidelity prototype using paper sketches. These sketches were later translated into a high-fidelity prototype implemented in JavaScript together with the React library, enabling interactive and dynamic functionality.

3.3 User evaluation and analysis

3.3.1 Participants. A total of six persons from diverse backgrounds participated in the study. Two of the participants that are the primary users of the current tool have military backgrounds where one is a combat pilot, and the other one an air combat controller. The other participants work with software

development at FLSC, meaning that they are technically proficient and have basic knowledge about military concepts.

3.3.2 Evaluation Setup. The six participants conducted the user tests and gave written consent to participate in the study. Before conducting the test, I described how the session would look like, where I said that I will give the user tasks to complete and that they should perform the tasks as how they think it should be done. I did not tell them that I was measuring learnability and error rate, as this could affect their behavior and cause them to try conducting tasks in a different way than they naturally would do it. The tests were both screen and audio recorded conducted on site at FOI in a meeting room with only the participant and me present in the room. The same computer was used for each test in order to minimize error caused by inconsistencies. The participant sat in front of the computer, and performed tasks based on verbal given instructions from me. During the entire session, I was screen and audio recording in order to provide material for the data analysis. After the user had conducted the tasks, an interview was conducted in order to gain further insights about the experience of the user interface. The recordings were then manually transcribed and acted as a foundation for the data analysis.

3.3.3 User tasks. The user tasks intended to capture behavior that is measurable and associated with learnability and error rate. The user tests consisted of 15 tasks that the participants would do along with a follow up interview with 10 questions afterwards. The tasks could be divided into the following general task types:

(1) Add a non-geometric entity
(2) Add a geometric entity
(3) Modify or access an entity's data
(4) Modify an entity's spatial position
(5) Add nested data

And were placed in the order 1, 2, 4, 1, 2, 3, 5, 4, 3, 1, 4, 4, 3, 4, 3. Similar task types were intentionally separated by other tasks rather than being placed consecutively. This was done to reduce immediate repetition effects and to allow learnability to be observed through participants recalling and reapplying interaction patterns after performing unrelated tasks.

3.3.4 Data analysis. The collected data consisted of quantitative measures (task completion time and error counts) and qualitative data from the interviews. The analysis aimed to identify associations between specific interface design choices and observed learnability and error rate outcomes.

3.3.5 Quantitative analysis. Learnability was measured by comparing task completion times within predefined task types. Each task set consisted of multiple tasks within the same task type (e.g., doing task 1 and then later doing task 1 again). The first task in a set represented initial interaction with the interface logic, while the later tasks reflected performance after prior exposure. For each participant, the completion times within the task sets were compared. A reduction in completion time between tasks with the same task type was interpreted as an indication of improved learnability of the interface. Error rate was measured by recording failed interaction attempts during task execution. Errors were logged at each procedural step (e.g., locating the correct button or menu, correctly selecting an element, entering data, saving the result). Both total error counts per task and step-specific error occurrences were analyzed. This allowed identification of interface elements or interaction stages associated with recurring confusion.

3.3.6 Qualitative analysis. The interview data was analyzed using qualitative content analysis. Statements related to specific interface features (e.g., color coding, menu placement, window management) were coded according to their evaluative stance (positive or negative). Recurring themes were identified across participants. Particular attention was given to statements referring to perceived ease of learning, clarity of navigation structure and experienced mental effort.

3.3.7 Integration of quantitative and qualitative findings. The final stage of the analysis involved linking the qualitative and quantitative data. A design choice was considered to be associated with high learnability when:

(1) Completion times decreased within a task set, and
(2) The majority of participants who discussed the design feature described it as easy to understand or reported that it became intuitive after initial use.

Similarly, a design choice was considered to be associated with low error rate when:

(1) Few errors were recorded in the interaction steps involving that feature, and
(2) The majority of participants who discussed the design feature described it as easy to locate, interpret, or use.

In this context, "few errors" refers to situations in which the majority of participants successfully completed the corresponding task. This approach allowed performance-based indicators and user-reported experiences to jointly inform conclusions about which design choices were most critical in supporting low error rate and high learnability.

4 RESULT

This section presents the results obtained during the design process of the prototype. The findings are based on demonstrations of the existing planning tool, interviews with operators, prototype development and user evaluations on the final end product.

4.1 Insights and desired features from operators

From the demonstrations and interviews with the operators of the primary tool, the core following insights were:

(1) **Lack of consistency**
The interface that the operators currently are using lacks consistency both in terms of other sub tools in the same systems, as well as other systems that the pilots use. Units where inconsistency appears are, for instance, distance units (meter versus foot), velocity (km/h versus knots) and coordinate system (UTM versus lat/long). This leads to increased cognitive load as the operators constantly have to make conversions themselves when switching between systems and tools. Moreover, it leads to increased errors since the conversions are not completely accurate. Compliance with military symbology standards is important as it reduces cognitive load [25].
(2) **Unwanted additional windows**
The operators stated that it was frustrating when multiple windows were created after clicking menus and buttons, resulting in a cluttered and messy interface. Previous research has shown that interfaces generating multiple windows from click events are associated with higher subjective workload and lower accuracy compared to interfaces designed according to Ecological Interface Design (EID) principles [10].

(3) **Lack of visual representations**
Lack of visual representation was furthermore expressed during the interviews. The ability to put different colors on polygons dependent on what team they are associated with, as well as displaying information when clicking on entities on the map were desired features. Visual feedback should be presented at the user's point of touch as they otherwise can find it difficult to carry out tasks in one location while attending to visual information on another location [4].

(4) **Insufficient error prevention**
The operators expressed frustration with the need to inspect disorganized log files in order to detect errors when creating or exporting route plans. They therefore requested improved error handling that could capture the user's attention earlier in the process, which would reduce both time and effort. In particular, the operators suggested that clear error messages, preferably highlighted with distinct colors, should appear when a user is about to perform an invalid action. This aligns with Nielsen's usability principles, which emphasize that error prevention can be supported by providing helpful constraints that guide users away from incorrect actions [17].

Next on, a meeting on site with the supervisor at FOI and the operators of the planning tool was held where the design requirements were defined. These requirements consisted of available operations that the tool should have, such as creating waypoints, having speed in the unit mach, creating polygons of different colors and displaying administrative boundaries on the map. Insights from the meeting with the operator from the visit to the flotilla consisted of inspiration in terms of system logic and user navigation logic. The material from this meeting consisted of written notes and five screenshots of the tool displayed on a computer.

4.2 Design

The final interface consisted of two main sections: a menu on the left and an interactive map on the right (see Figure 2). The menu featured a tab panel with three categories: route, areas, and aircraft (see Figure 3). Each tab followed a consistent layout, consisting of a list of entities on the left and a corresponding details panel on the right, displaying information about the selected unit. At the bottom of the details panel, an edit button enabled users to modify the attributes of the selected unit. The aircraft tab differed slightly from the others by including an additional "create formation" button within the entity list (see Figure 7). This functionality was exclusive to aircraft, as they are the only units that can be grouped and displayed as nested elements within the list. Below the tab panel and lists, a set of action buttons was provided. Each button represented a specific function, indicated by its icon, and some buttons expanded into dropdown menus when selected as shown in Figure 4. For example, selecting the "W" button enabled waypoint placement by allowing users to click directly on the map. Similarly, selecting the "L" button opened a dropdown menu with two lists, enabling users to choose start and end landing bases, which were then automatically placed on the map. Similar to the landing base menu, other menus and actions were restricted when required data had not been provided, thereby minimizing potential user errors. The geometric units and aircraft also included dropdown menus for color selection, allowing users to create shapes and aircraft in different colors. All units displayed on the map were interactive, meaning in this case that a popup menu appeared when clicking on the unit, allowing for faster lookups or delete actions (See Figure 5). Selecting a unit on the map would highlight the corresponding entry in both the entity list and the details panel, where all units could be edited (see Figure 6). Additionally, units could be repositioned via drag-and-drop interactions or removed through direct interaction. The map itself

included predefined territorial and administrative boundaries and was styled using a light grayscale color scheme, with the intention of minimizing visual clutter and thereby reducing cognitive workload for the user [6]. Furthermore, graticules, landing bases and their corresponding airspaces and track directions was included in order to support users´ awareness of the surrounding. An import and export button were placed at the bottom right corner of the menu panel, supporting the use of both exporting scenarios and uploading scenarios. If no units were placed out, the export button would be disabled as to hinder unecessary exports of scenarios that does not contain anything. Similarly, the import button would get disabled if a scenario already was started.

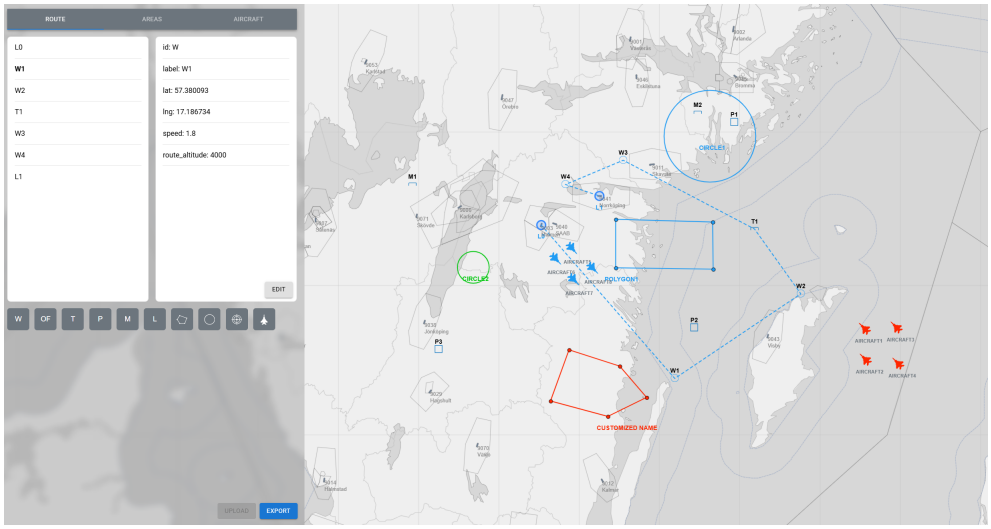


Figure 2: The interface with the menu to the left and the map to the right



Figure 3: The tab menu, having the blue underscore indicating which category the user currently is positioned on

4.3 Evaluation with users

The following chapter will go through the user tests that covered the tasks and interviews.

4.3.1 Quantitative analysis. For task type 1, all participants demonstrated a decrease in task completion time across repeated tasks. These tasks included adding waypoints, points, and landing bases. For waypoints and points, all participants successfully navigated to the toolbar, selected the correct tool, and placed the object on the map with a single left click. For landing bases, 5 out of 6 participants successfully selected the correct option, chose airbases from the dropdown menu, and confirmed placement by clicking the “OK” button (see Figure 4).

For task type 2, participants were asked to create geometric shapes, specifically a blue polygon followed by a red circle. The first step in both tasks required navigating to the correct tool, selecting it, and

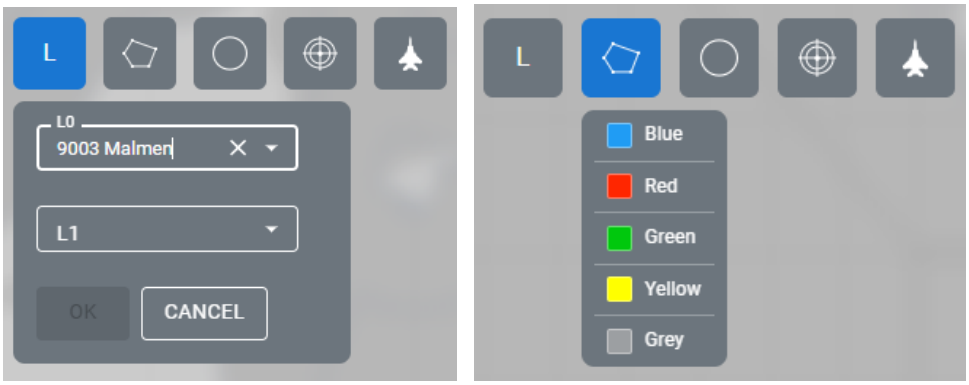


Figure 4: Extendable menus for certain buttons



Figure 5: Popup boxes when right clicking on units, where aircraft has an additional rotate symbol

choosing a color from an extended menu (Figure 4). The interaction process then differed depending on the shape. Creating a polygon required placing vertices through successive left clicks, after which the shape could be completed by clicking the first vertex, right-clicking, or reselecting the polygon tool. The current vertex to be placed out was connected to the first point as well as the former vertex in order to give the user an idea of how the polygon may look like, along with a cross to symbolize that the user is in place mode and can place the vertex where the cursor hovers (see Figure 8).

Creating a geometric figure required first left-clicking on the map to either create the center for the circle or to create the first vertex in a polygon. In order to finalize the circle, the user had to drag the cursor in order to define the radius and then left-click again to finalize the circle. All participants showed a reduction in completion time for the initial step of selecting the correct tool and color. However, difficulties emerged during shape creation. Five out of six participants successfully placed the first polygon vertex, while one participant failed to select a color beforehand, causing the tool to deactivate. To complete the creation of the polygon on the map, half of the participants initially used incorrect interaction methods before identifying the correct procedure. For example, two participants attempted to complete the polygon using the Esc key, while one participant attempted to finalize it through double-clicking. For the circle task, two participants initially attempted to define the radius by dragging immediately upon selecting the tool, without first placing the center point. After correctly placing the center point, 5 out of 6 participants successfully completed the circle with a left-click, while one participant attempted to use a right-click instead.

Task type 3 involved deleting units, looking up information about them and renaming them. All participants successfully completed these tasks without any failed attempts. A reduction in completion time

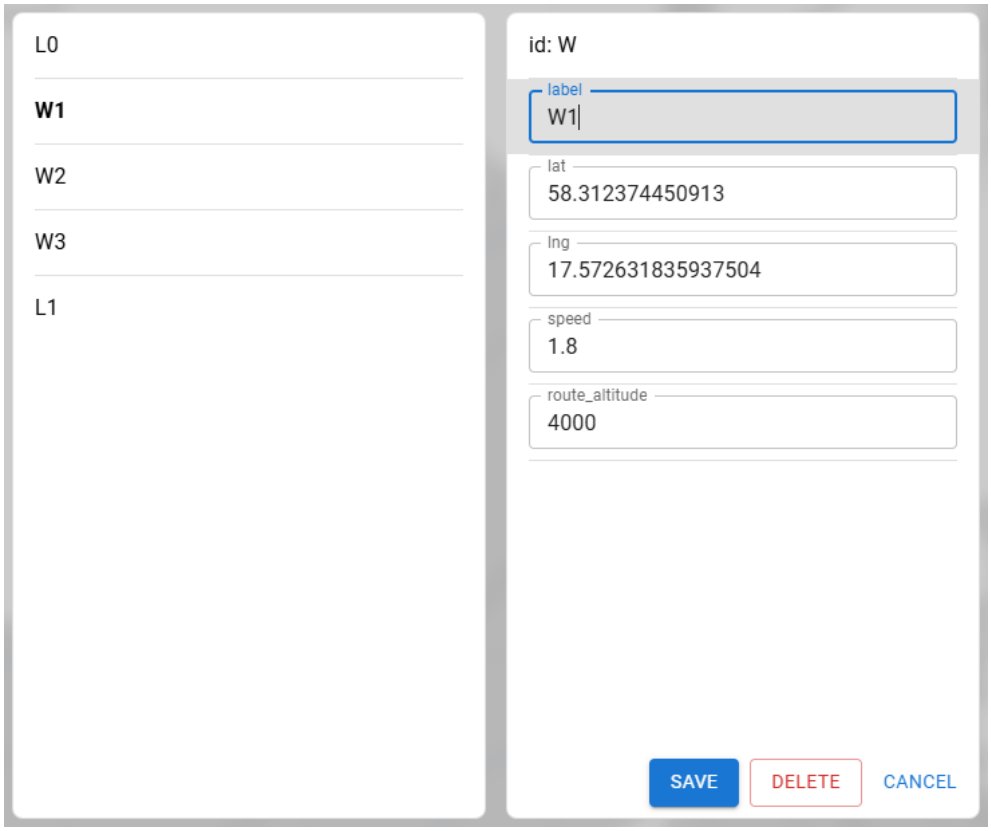


Figure 6: The edit menu in details list

was observed for 5 out of 6 participants. The participant who did not show any negative difference in time and therefore improvement was the only one who immediately used the menu, whereas the others initially interacted with the units directly on the map to access the corresponding menu options.

Task type 4 involved moving units geographically and adjusting their rotation. All participants successfully repositioned points and circles. However, only 4 out of 6 participants successfully changed the rotation of an aircraft. Among those who succeeded, two participants performed the action by selecting the aircraft and dragging the rotation handle that appeared (see Figure 5), while the other two used the rotation controls in the details menu. The remaining two participants attempted to rotate the aircraft by dragging it directly, suggesting an expectation that rotation would be available through direct manipulation without entering a specific rotation mode.

Task type 5 involved creating aircraft formations, which was done within the list of displayed aircraft (see Figure 7). Three participants successfully completed this task. The remaining participants attempted alternative approaches, such as selecting multiple aircraft by dragging with the right mouse button or using keyboard modifiers to select multiple units. A common observation across all participants was initial hesitation after selecting the aircraft tool, before placing aircraft on the map.

4.3.2 Qualitative analysis. All participants expressed an overall positive first impression of the interface. It was described as *"modern"*, *"intuitive"* and *"something one wants to work in"*. Three participants noted that the system aligned well with their mental model of how tasks should be performed,

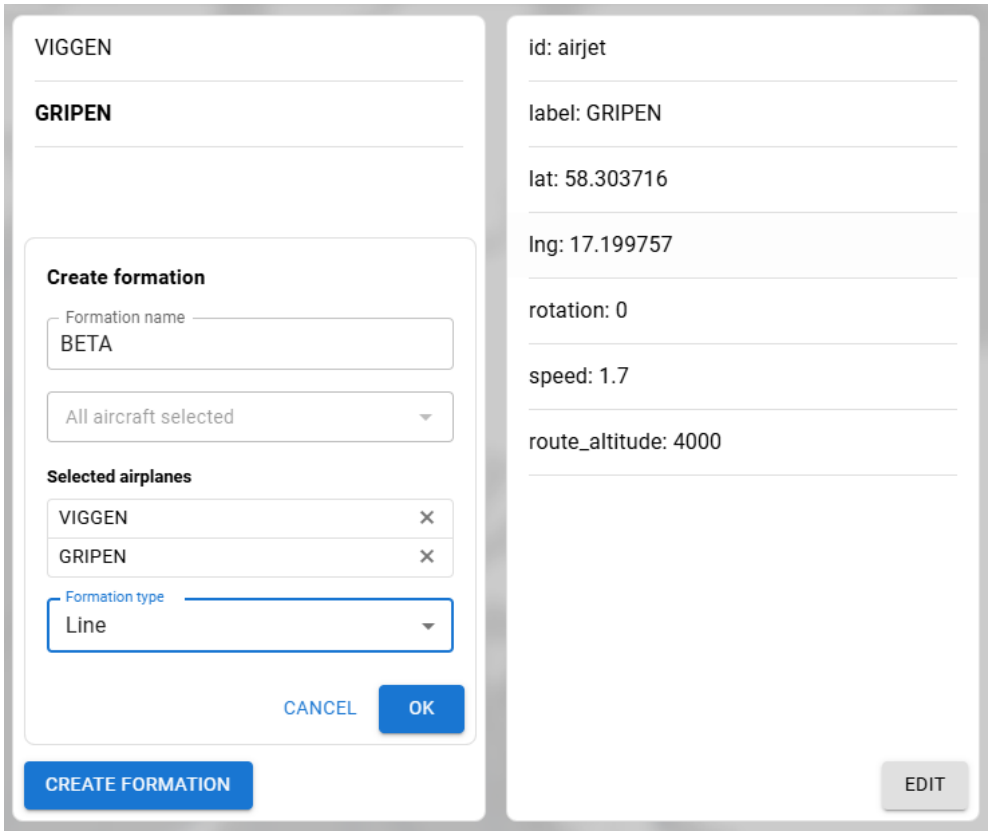


Figure 7: The formation menu inside the informationlist for aircraft

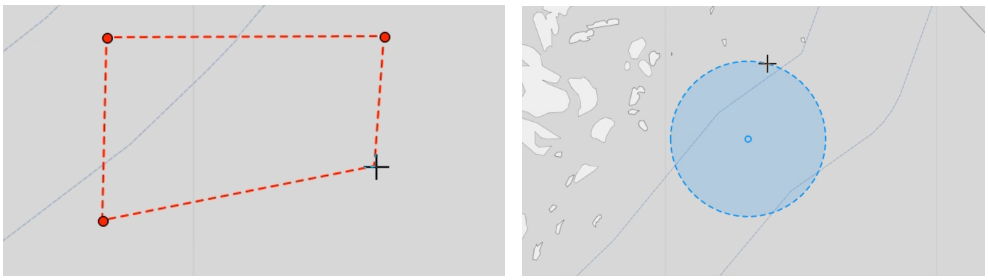


Figure 8: Creation of geometric figures

particularly in relation to object-based interaction where they expressed that the system felt closer to the user rather than the computer. At the same time, some aspects of the interface were perceived as less clear. In particular, the left-hand menu, including its buttons and symbols, was described by two participants as somewhat cryptic and difficult to interpret initially. However, one participant who frequently used the menu reported that they quickly learned its structure and logic, and another participant described the symbology of the interface as intuitive. Regarding perceived ease of use, all participants rated the interface between 8 and 10 on a 10-point scale.

When asked whether participants learned anything during the user test that made later tasks easier, the responses varied. Three participants referred to the recognition of recurring task patterns. One mentioned the flow of changing names on units, and another one mentioned the pattern of creating geometric figures, saying:

> Multiple buttons remind of each other, and I understood that it would follow the same flow when choosing a colored polygon as with the colored circle earlier.

Another participant mentioned understanding that the menu dynamically updates to reflect changes made on the map. The remaining two participants pointed to learning to create and manipulate circles, which contributed to making the interface easier to learn and use. On the question if the participant experienced that it was clear how to navigate on the interface, one participant responded *"It was easy, as things that had to do with each other were on the same place in the interface"*. In addition, four participants mentioned that the tasks they initially found difficult became easy once they understood how the interaction should be performed. When asked whether there was anything in the interface they misunderstood, one participant reported confusion when switching between selected units for new placements. The participant explained that this confusion came from being accustomed to how this interaction is handled in the current system.

Several tasks were consistently described as easy to perform. These included placing units belonging to the route, adding aircraft, and moving or rotating units. These interactions were perceived as intuitive, largely due to their reliance on direct manipulation and consistent interaction patterns between different elements. At the same time, certain tasks were identified as more challenging. In particular, creating formations and working with geometric shapes (such as circles and polygons) caused difficulties for multiple participants. One participant reported struggling with formation creation because they were unable to locate the relevant button. Another participant reported difficulty in creating the formation, as their attention centered primarily on the map. As a result, they did not notice the menu on the left where the formation function was located. Difficulties related to geometric shapes were often attributed to prior experience with other systems, where different interaction patterns are used to create such elements.

Participants with a military background expressed particular appreciation for the domain-specific functionality. For example, the availability of a predefined list of airbases, which could be placed directly on the map, was seen as a significant improvement compared to manually entering airbase codes or positioning them geographically. Additionally, one participant highlighted the built-in error prevention, noting that the inability to move airbases on the map was a valuable feature, as this prevents inaccuracies.

4.3.3 Integration of quantitative and qualitative findings. Across all types of units, participants showed a decrease in the time required to locate and select the correct tool for placing elements on the map. Although two participants initially described the menu bar as cryptic, the overall reduction in completion time indicates that having clearly separated and persistently visible action buttons supports learnability and reduces errors over time.

A similar pattern was observed for geometric figures. Participants demonstrated reduced completion times when selecting the correct tool and color, and qualitative feedback highlighted that these features followed a consistent interaction logic. This suggests that reusing the same interaction patterns across similar elements facilitates learning and enables users to apply previously acquired knowledge, thereby supporting high learnability. Since five out of six participants selected the figure tool before choosing a color, the interaction flow appeared to align well with user expectations.

All participants successfully moved units by dragging them directly on the map, and accessed deletion functionality by left clicking on a unit and using the corresponding popup interface control. Moving, accessing information and deleting units were also described as easy tasks by participants, indicating that these implementations, signified by direct manipulation, are associated with low error rates.

Changing the name of units also showed reduced completion times across repeated tasks, and was consistently described as easy to perform. This indicates that consistent interaction patterns for editing attributes available on the same spot by an edit button in the interface support high learnability. Positioning the edit button within the details list, where the corresponding information is presented, appears to be a design choice associated with low error rate.

The results suggest that when interactions related to spatial or geometric tasks are not performed directly within the map, users may struggle to understand how to complete them, leading to hesitation and lower task success. Similarly, the observed difficulties in completing geometric shapes, together with participants' references to prior experience with other systems, suggest that deviations from familiar interaction patterns may lead to increased uncertainty and incorrect attempts.

5 DISCUSSION

This section discussed the method and results in order to answer the research question: *Which design choices are most critical for supporting low error rate and high learnability in a digital map-based planning tool for air combat scenarios?*

5.1 Learnability and error rate as metrics

In this study, learnability was primarily reflected in the fact that a total of between 4-6 participants completed tasks within each task set with decreasing completion times. This trend suggests that users were able to quickly adapt to the structure and interaction patterns of the interface. In addition, participants expressed positive feedback regarding the corresponding design elements, indicating that these elements supported a consistent and predictable interaction logic.

At the same time, many tasks took relatively little time to complete even during the first attempt, resulting in only marginal differences between initial and subsequent task completion times. Rather than solely reflecting incremental learning, this may indicate that certain interaction patterns were already familiar to users, potentially due to prior experience with similar systems or established design conventions.

However, such familiarity should not necessarily be interpreted as evidence of effective design. The fact that users quickly understand or adopt a particular interaction pattern may instead reflect prior exposure rather than the inherent quality of the design itself. In other words, a design can appear to have high learnability simply because it aligns with conventions that users are accustomed to, even if those conventions are suboptimal. A similar limitation applies to error rates, where low error frequency may not always indicate that a design is intuitive, but rather that users rely on ingrained habits or expectations when interacting with the system.

Low error rates do not necessarily mean that users immediately understood where to interact within the interface. In some cases, correct actions may have resulted from guessing or trial-and-error, rather than from confident decision-making. Although four of six participants reported that it was clear where to click for different tasks, this retrospective feedback may not fully capture moments of hesitation or uncertainty during task execution.

Four participants reported that their difficulties with certain tasks were related to being accustomed to performing them differently. These findings are consistent with previous research suggesting that interfaces that are inconsistent with user expectations may increase the likelihood of human error [11]. At the same time, the four out of the six participants that stated that it was generally clear where to click in the interface did occasionally select incorrect elements during the tasks. This suggests that perceived clarity does not necessarily correspond to correct interaction. Users may rely on prior experience and established habits, leading them to make incorrect selections despite reporting that the interface feels intuitive. As a result, error rates in this study may partly reflect mismatches between the interface and the users' previous experience, rather than purely indicating unclear design. This suggests that error rate is most meaningful when interpreted together with qualitative feedback and observations of user behavior, rather than as an isolated metric.

This suggests that both learnability and error rate, as applied in this study, may overlook more subtle aspects of user experience, such as hesitation, doubt, or reliance on trial-and-error strategies. Consequently, these metrics should be interpreted with caution, as they may overestimate how intuitive or self-explanatory the interface actually is. To capture these aspects more accurately, it may have been beneficial to collect immediate feedback after each task, for example, by asking participants whether they felt confident in their actions or experienced uncertainty when deciding where to click. Such an approach could have provided a more fine-grained understanding of how users perceived and interacted with the interface in real time.

5.2 Prototype

Due to time limitation, all desired functionality could not be implemented in the prototype, which lead to limited amount of functionality that could be tested. An alternative solution that would cover more features in the interface would perhaps be to use a tool like Figma. However, due to the nature of the tasks, the prototype needed to support all kinds of placements and adjustments of units which would have been very difficult in a tool like Figma.

Moreover, the screen and audio recordings along with the running web server happened to do the website slower and bugging which affected some user tests, where parts of the map would disappear, such as county lines for instance. They came back as soon as the user interacted on the screen again and did affect the functionality. However, its behavior might have bothered the users and subconsciously affected their effort in completing tasks.

Finally, two participants referred to functionalities that were affected by bugs when asked whether anything in the interface was unclear. For example, one participant encountered a bug where it was not possible to create a polygon inside a circle. This issue was resolved after that participant's session and before the remaining user tests, as this functionality was intended to be supported. As a result, the participant's experience may have been disproportionately influenced by this issue. If the bug had been resolved beforehand, it is possible that the participant would instead have thought about other limitations or challenges in the interface for the interview after.

5.3 Participants

This study included a relatively small sample of six participants, of which only two belonged to the intended user group, consisting of air combat pilots and air combat controllers. As a result, the generalization of the findings is limited, as the majority of participants may not fully represent the knowledge, experience, and expectations of the target users. In some cases, the absence of domain-specific knowledge affected task performance. For example, one participant expressed uncertainty regarding the meaning of "L0" and "L1", which led to increased completion time and required clarification during task execution. This suggests that some observed difficulties may be attributed to unfamiliarity with domain concepts rather than issues related to the interface itself.

At the same time, it is possible that similar uncertainties occurred without being explicitly verbalized by participants. In such cases, correct task completion may have resulted from guessing rather than understanding. Conversely, certain errors may also have been caused by a lack of domain knowledge rather than unclear interface design.

Moreover, the two participants with a military background, who also represent the intended end users, were involved during the development phase of the prototype in order to provide guidance and feedback. This prior exposure may have influenced their performance during the evaluation, as they could have become somewhat familiar with aspects of the interface. However, their involvement was limited to the map component of the prototype, and they did not interact with the menu interface beforehand as this was not implemented when they saw the interface. Therefore, the interactive parts of the map were still new to them during the evaluation.

Overall, this indicates that both performance measures and error rates in this study may be influenced by participants' varying levels of domain expertise as well as their involvement during the design phase, which should be taken into account when interpreting the results.

5.4 Evaluation setup

Due to hardware limitations of the computer provided by FOI, it was not possible to record screen and audio simultaneously. Instead, screen recordings and audio recordings had to be captured separately and manually synchronized afterwards by using an audible cue at the start of each session. As a result, there may be minor inconsistencies regarding the exact synchronization between the audio and screen recordings. Because photography and video recording were strictly prohibited within the FOI facility, it was not possible to use an external device, such as a phone camera, to obtain fully synchronized audio and video recordings.

Tracked time completion would perhaps be more accurate if I would have been tracking each task with a stopwatch and then transcribing the completion times instead of calculating the times based on the screen and audio recordings. This was considered, but I concluded that this approach probably would contain human errors as well, such as forgetting to start or stop timestamp on certain tasks. This would probably be a problem in those cases that users asked questions during tasks.

This thesis does not include detailed tables of task completion times within each task type. This decision was made due to technical challenges during data collection, which may have introduced minor inaccuracies in the recorded times. Presenting exact time differences would therefore risk implying a level of precision not fully supported by the data. Within the scope of this thesis, the observed trend of decreasing completion times across repeated tasks is considered more relevant than the exact numerical differences, as it sufficiently indicates improvements in learnability. The delay between the audio

and screen recordings was estimated to be at most 1.5 seconds. Consequently, only reductions in task completion time exceeding this margin were considered reliable indicators of improved performance.

Furthermore, it would be valuable to examine task completion times and error rates over a longer period rather than within a single evaluation session. For example, the study could be structured into multiple sessions conducted on separate days. Such a setup would allow for the assessment of how well users retain knowledge of the interface over time, rather than only capturing short-term learning effects. This approach could provide more meaningful insights into the system's learnability and usability, as it better reflects how the system would be used in practice.

The evaluation part consisted of predefined tasks that I gave oral instructions to the participants to complete, followed by an interview afterwards. Reflecting over the difficulty of the tasks, it is clear the more advanced tasks such as creating geometric figures and formations should have occurred multiple times. These tasks had the highest error rates but did not occur multiple times and therefore could give a sense of detected learnability.

For participants outside the intended user group, performance may also have been affected by a lack of domain-specific knowledge. Providing a brief introduction to key concepts used in the evaluation, such as the meaning of "L0" and "L1", could have reduced this source of uncertainty and allowed for a clearer assessment of the interface itself.

Completing tasks while being observed may have been perceived as a stressful and uncomfortable situation, as participants might have felt uneasy making mistakes or not understanding certain elements of the interface. An alternative setup, where I was not physically present, could have created a more relaxed environment and potentially allowed participants to perform the tasks with greater comfort and therefore potentially other outcomes. At the same time, my presence during the evaluation may have influenced how honestly participants expressed their opinions about the interface. If the interviews had been conducted by someone not involved in the development of the prototype, participants might have felt more comfortable providing critical feedback, as they would not need to consider the interviewer's personal involvement in the design.

Finally, reliance on post-task interviews introduces a risk of recall bias, as participants may not accurately remember moments of hesitation or confusion. Collecting feedback directly after each task could have provided more immediate and detailed insights into user decision-making during interaction.

5.5 Design implications

Despite the limitations discussed, several patterns can be identified regarding which design choices appear to support efficient interaction in the interface. One of the most prominent findings is the importance of consistent interaction patterns. Participants were able to transfer knowledge between tasks, as similar actions followed the same structure across different elements. In practice this means having a color drop down menu when selecting buttons for creation of geometric figures, having units belonging to the route be placed out by single left clicks on the map, separating edit actions to a left side menu regardless of the unit type and using a constantly visible menu for clicks and then placing them out. This was reflected in decreased completion times together with few errors and supported by participants' feedback, indicating that consistency allows users to quickly and accurately understand and predict how the interface behaves. The observed transfer of knowledge between tasks supports previous findings where the authors imply that consistent interaction patterns reduce cognitive workload and allow users to focus on task execution rather than interface interpretation ([11], [14]).

Another important aspect was the use of direct manipulation within the map interface. Tasks involving placing, moving, deleting or interacting with objects directly on the map were consistently completed with few errors and were described as intuitive by participants. Just as with the study by [19], participants preferred a drag-and-drop approach for moving units instead of using the button which would let them type in the new coordinates in the menu bar. The same pattern was observed when deleting or modifying the data of the unit, as the majority of the users clicked on the unit directly instead of using the details view. These interactions, along with most other interface actions, provided visual feedback at the user's point of interaction. This finding aligns with [4], who argue that feedback should be presented at the user's point of touch. This suggests that allowing users to interact directly with visual elements, rather than relying on indirect controls, supports low error rate. This does furthermore align with the related previous research, whose authors suggest that direct manipulation can improve learnability and allow novice users to learn basic functionality rapidly ([22], [20]).

In contrast, more complex interactions, such as creating geometric shapes and formations, resulted in higher error rates and user uncertainty. These tasks often required multi-step interactions or less visible controls, which made them harder to discover and execute. This indicates that complex functionality may require additional design support, such as clearer guidance, more explicit feedback, or step-by-step interaction cues.

Overall, the findings together with previous research indicate that design choices that emphasize consistency, and direct interaction support high learnability and low error rates. However, more complex features require additional attention in order to maintain these qualities across the interface.

5.6 Ethics and sustainability

The thesis is conducted within a military organization, which may raise ethical considerations related to activities involving simulated combat. However, the work focuses on interface design and planning tools rather than operational or tactical decision-making. From a human factors perspective, improving learnability and reducing interaction errors in planning systems may also contribute to safer and more efficient training environments. Furthermore, interfaces that are consistent and therefore easier to understand can reduce cognitive workload and decrease the likelihood of user mistakes during planning activities. From a sustainability perspective, the increased use of virtual training environments reduces the need for live flight exercises, thereby lowering fuel consumption and associated emissions compared to training with real aircraft [2].

Generative AI has occasionally been used in order to improve phrasing and wording in this thesis.

5.7 Limitations

The findings of this study should be interpreted in light of several limitations. First, the small and partly non-representative sample limits the generalizability of the results. Since most participants did not belong to the intended user group, their interaction patterns and error rates may not fully reflect those of experienced air combat planners. As a result, some observed difficulties may be related to differences in domain knowledge rather than the interface design itself.

Second, the limited scope of the prototype means that only a subset of the intended functionality was evaluated. This may have influenced both learnability and error rates, as a more feature-complete system could introduce additional complexity and thereby affect user performance. Furthermore, the evaluation

was conducted within a single session, capturing primarily short-term learning effects. This limits the ability to assess how well users retain knowledge of the interface over time.

Although the findings are exploratory due to the limited sample size, several consistent patterns emerged across participants, particularly regarding direct manipulation and consistent interaction logic.

5.8 Future research

For future research, it would perhaps provide more accurate information if the users were given the task to plan a full scenario and therefore not getting predefined tasks in order to see how they use the prototype in a more natural setting. For instance, one of the participants with military background asked when setting out waypoints ”should I do that before setting out L0 and L1?” indicating that the flow of the tasks were a bit unnatural. Since some participants in this study lacked knowledge of how to plan a scenario, it could not be done in this study and instead I had to dictate what they should do in order to test how they respond with the different elements of the interface.

Moreover, it would be interesting to test on a more fully developed interface in order to get more valuable insights. This study had a prototype with a limited amount of implemented features. This might have perhaps increased the learnability and decreased the error rate, as there are not that many options to choose from that makes the interface difficult to navigate in or get lost in. A more fully developed interface would also be closer to reality and therefore give more reliable results perhaps.

As noted above, it would be valuable to measure learnability and error rates over a longer period, such as across one or more days. This approach does not necessarily need to replace the evaluation setup used in this study, but could instead complement it. Such an approach would provide a more comprehensive understanding of both the system’s short-term and long-term usability and learnability.

More participants with adequate background would be interesting to test on in a future study in order to be able to draw broader conclusions. This would reduce the influence of prior habits or shortcuts from other systems, thereby increasing the objectivity of the study.

6 CONCLUSION

This study set out to explore which interface design choices are associated with high learnability and low error rates in a map-based military planning tool. The results indicate that consistent interaction patterns and direct manipulation of map elements are key factors in supporting both rapid learning and low error rates. Users were able to transfer knowledge between tasks when similar actions followed the same structure, which reduced both completion time and interaction errors.

In contrast, more complex interactions that relied on less visible controls or multi-step processes were associated with higher error rates and increased user uncertainty. These findings highlight the importance of visibility, consistency, and alignment with user expectations when designing interfaces for complex systems.

The study also shows that prior experience influences both performance and perceived usability. As a result, metrics such as error rate may not always reflect true understanding, but rather the extent to which the interface aligns with users’ previous habits.

Overall, the findings contribute practical design guidelines for improving usability in map-based planning tools and similar systems, where efficient interaction and low error rates are critical.

7 ACKNOWLEDGMENTS

I would like to thank FOI for giving me the opportunity to conduct this thesis project with you. I am especially grateful to my supervisors Rickard Fahlquist and Tina Karlsson for their continuous support, encouragement, and genuine willingness to help throughout the project. I would also like to thank my academic supervisor Henrik Artman for his valuable feedback and insightful discussions during the work.

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